

Both components of the wind loads shall be applied as line loads at the mid-depth of the superstructure. In plan, the longitudinal components of wind loads shall be applied as line loads along the longitudinal axis of the superstructure.

The skew angle shall be taken as measured from the perpendicular to the longitudinal axis of the bridge in plan.

The wind direction for design shall be that which produces the maximum force effect in the substructure. The transverse and longitudinal wind load components on the superstructure shall be applied simultaneously.

For girder bridges, the wind pressure may be taken as one line load whose intensity is equal to the product of the wind pressure, skew coefficients, and the depth of the superstructure including the depth of the girders, deck, floor system, railing, and sound barriers, as seen in elevation perpendicular to the longitudinal axis of the bridge. For trusses, columns, and arches, the wind load is the sum of the wind loads on the windward and leeward areas.

The purpose of applying the line load along the longitudinal axis of the bridge in plan is to avoid introducing a moment in the horizontal plane of the superstructure.

Table 3.8.1.2.3a-1—Skew Coefficients for Various Skew Angles of Attack

Skew Angle (degree)	Trusses, Columns, and Arches		Girders	
	Transverse Skew Coefficient	Longitudinal Skew Coefficient	Transverse Skew Coefficient	Longitudinal Skew Coefficient
0	1.000	0.000	1.000	0.000
15	0.933	0.160	0.880	0.120
30	0.867	0.373	0.820	0.240
45	0.627	0.547	0.660	0.320
60	0.320	0.667	0.340	0.380

For usual girder and slab bridges having an individual span length of not more than 150 ft and a maximum height of 33.0 ft above low ground or water level, the following wind load components may be used:

- Transverse: 100 percent of the wind load calculated based on wind direction perpendicular to the longitudinal axis of the bridge.
- Longitudinal: 25 percent of the transverse load.

Both forces shall be applied simultaneously.

3.8.1.2.3b—Loads Applied Directly to the Substructure

The transverse and longitudinal forces to be applied directly to the substructure shall be calculated using the wind pressure determined using Eq. 3.8.1.2.1-1. For wind directions taken skewed to the substructure, the wind pressure shall be resolved into components perpendicular to the end and front elevations of the substructure. The component perpendicular to the end elevation shall act on the exposed substructure area as seen in end elevation, and the component perpendicular to the front elevation shall act on the exposed substructure area as seen in front elevation. The two substructure wind force components shall be applied simultaneously with the wind loads from the superstructure.

C3.8.1.2.3b

When combining the wind forces applied directly to the substructure with the wind forces transmitted to the substructure from the superstructure, all wind forces should correspond to wind blowing from the same direction.